Design Requirement Justification/Research

Priority	Design Requirements	Reason	Rationale
High	**Measures upper, middle, and lower chest expansion**	The garment shall measure expansion in the upper, middle, and lower chest for toddlers wearing the device with chest circumference, taking the circumference at the end of exhalation and at the top of inhalation. This will be shown using the absolute values of both and the difference of the two values. The expansion will be measured in centimeters to the nearest millimeter.	Dr. Brossman emphasized the need for measuring chest expansion in multiple areas of the chest to establish normative data. Email History
High	**Durability** **(Must hold its shape, hold the sensors in place, and be able to stretch back and forth)**	The garment must be capable of holding its shape and its sensors in place after numerous uses and washing machine cycles. It must also be able to stretch and revert back to its original form without damaging the fabric or any of the sensors. Having minimal movement could also help produce more accurate results. (Measurement is not moving throughout the onesie while a breath is happening) • Sensor holders should be within 1cm of their original placements	The onesie needs to be able to hold its shape to minimize the costs of replacement while also ensuring it is readily available for PT assessments. This is to prevent the baby from getting any infection or illness.(Refer to Zoom Meeting doc)
High	**Safe**	The device shall not injure or restrict the functions of the toddler in any way. The toddler must be fully safe and not negatively affected by the garment. This means the garment cannot overheat or force the body of the toddler to bear excessive load. Putting the toddler in an	The toddler needs to be totally unharmed (Refer to Zoom Meeting doc)

		unsafe environment could also lead to inaccurate test results. ● There should be no sharp edges where the subject can be injured.	
Low	**Measures breathing rate**	The garment shall measure breathing rate for toddlers wearing the device in breaths per minute, measured by the frequency of inhalation and exhalation, while ensuring that the baby is not affected negatively in any way by the device.	Dr. Brossman would like for the device to measure breathing rate, but she primarily wants chest expansion Email History and Zoom Meeting
Low	**Measures heart rate**	The garment shall measure the heart rate of the toddler in beats per minute by a sensor that keeps track of BPM, while ensuring the toddler is safe and not negatively affected. This will be measured by tracking the average temperature that all sensors touching the toddler.	Dr. Brossman would like for the device to measure heart rate, but she primarily wants chest expansion Email History and Zoom Meeting
Low	**Data sent to an application**	The garment should be able to send its data to an application that allows easy access. Analyzing the data through a computer or phone app is accessible for almost anyone and could also allow for more accurate results. Real time data could also be very helpful to compare each breath to the toddler in real time within ½ minutes.	StretchSense Harvard research (They no longer sell it) (Check email history extra tab for letter) stretchsense